

VibeFilming: An End-to-End Autonomous Director Agent Framework for Complex Video Creation Tasks

Abstract

Current AI video generation tools are predominantly confined to single-shot generation or fragmented workflows, rendering them inadequate for complex video creation tasks that demand narrative planning and multi-shot coordination. This paper proposes VibeFilming, an end-to-end autonomous director agent framework that conceptualizes video creation as a planable, executable, and rectifiable long-horizon task. The framework adopts a five-stage pipeline architecture comprising five collaborative agents: screenwriter, storyboard, visual generation, quality control, and director. Through artifact provenance tracking and reflection mechanisms, the system enables precise error localization and localized repair. Experimental results demonstrate that VibeFilming significantly outperforms baseline methods in narrative coherence and visual consistency, offering a viable pathway for AI video creation to evolve from stochastic “shot generation” toward systematic “industrial-grade production.”

Keywords: Video Generation; Multi-Agent Systems; Large Language Models; Autonomous Directing; Narrative Planning

1 Introduction

Video creation constitutes one of the most intellectually demanding forms of human creative labor. From the initial conception of a narrative premise and the development of a full screenplay, through the meticulous design of storyboards and character aesthetics, to the intricate choreography of camera movements and the nuanced artistry of post-production editing, every phase demands not only deep domain expertise but also refined aesthetic sensibilities that have traditionally required years of professional training to cultivate. Moreover, these stages are bound together by dense causal dependencies—a weak premise undermines the entire narrative architecture, while an imprecise storyboard inevitably leads to visual confusion during assembly. Traditional film and television production relies on linear collaboration across dozens of specialized roles, encompassing screenwriters, directors, cinematographers, production designers, editors, and sound engineers, entailing extended production cycles, prohibitively high costs, and steep barriers to entry. These factors have historically confined professional-grade video content production to a limited number of well-resourced institutions, leaving the vast majority of creative individuals without access to professional production capabilities.

In recent years, the rapid evolution of deep learning, particularly the breakthrough of diffusion models, has dramatically transformed the landscape of visual content creation. Text-to-image generation has matured to the point where photorealistic images

can be synthesized from natural language descriptions with remarkable fidelity. Building upon this foundation, video generation models have made equally impressive strides, with technologies represented by Stable Video Diffusion, Sora, and domestic systems such as Seedance and Kling now capable of producing high-fidelity dynamic footage within seconds from simple textual prompts. These advances have democratized visual content production to an unprecedented degree, enabling individuals without any technical or artistic training to generate striking visual material.

Nevertheless, a fundamental gap persists. The vast majority of these models operate within what may be characterized as the “single-instruction, single-shot” paradigm—users provide a prompt, and the model returns a short, isolated video clip, typically lasting no more than a few seconds. While this lottery-style generation can occasionally produce visually stunning individual shots, it remains fundamentally incapable of addressing a far more consequential question: how can AI be orchestrated to produce complete, narratively coherent videos that tell meaningful stories across multiple shots and scenes, with sustained character identities, consistent visual styles, and logically flowing sequences of events? The broader AI community has increasingly reached consensus that the next frontier for AI video creation is no longer the isolated performance metrics of individual generation models, but rather the systematic integration of distributed capabilities into cohesive, end-to-end intelligent systems capable of managing the full complexity of long-form narrative production. This paper directly responds to this pressing demand by proposing the VibeFilming framework, articulating a comprehensive technical pathway through which AI systems can autonomously accomplish complex, multi-shot video creation tasks, thereby elevating the prevailing paradigm from instantaneous prompt-response to deliberative planning and systematic execution.

2 Problem Formulation and Challenge Analysis

2.1 Task Characteristics of Complex Video Creation

Within the scope of this paper, “complex video creation” is formally defined as narrative-driven video production that incorporates multiple shots and scene transitions, with a typical duration exceeding thirty seconds. This category of tasks is fundamentally distinct from short-form template-based filling or isolated single-shot generation, and is characterized by three core attributes that collectively define its essential complexity.

The first attribute is long-horizon planning dependency. Unlike single-shot generation, which requires only local coherence within a brief temporal window, complex video creation demands global coordination across potentially hundreds of individual shots. The creator must maintain narrative logic, character developmental arcs, emotional pacing, and rhythmic variation throughout the entire duration, while simultaneously honoring multi-layered causal and temporal constraints that link distant segments of the work. This imposes demands on AI’s long-term reasoning and strategic planning capabilities that far exceed the cognitive requirements of single-turn generation tasks, pushing the boundaries of what current large language models can reliably achieve when embedded in production systems.

The second attribute is multi-modal asset coordination. Complex video creation necessitates the concurrent management and temporal alignment of heterogeneous asset types, including textual scripts, visual character designs, scene backgrounds and environments, audio dialogue tracks, sound effects, and musical scores. Each of these modalities oper-

ates according to its own structural logic, yet all must be precisely synchronized along the shared temporal axis to produce a coherent final product. Discordance or misalignment in any single modality risks compromising the perceived integrity of the entire work, making cross-modal consistency a particularly vexing challenge for automated systems.

The third and perhaps most operationally significant attribute is error propagation and accumulation. In multi-stage creative workflows, imperfections introduced at upstream stages—whether a weakly motivated plot point in the script, an ambiguous stage direction, or an imprecise camera specification in the storyboard—tend not to remain isolated. Instead, they are systematically amplified as they propagate through subsequent generation stages, manifesting ultimately as global narrative fractures, jarring visual discontinuities, or irreconcilable character inconsistencies. This means that downstream stages cannot effectively compensate for upstream deficiencies through localized adjustments or repeated generation attempts; rather, the integrity of the final product depends critically on the quality and precision of the earliest planning decisions.

2.2 Limitations of Existing Methods

Despite the remarkable progress in underlying generation technologies, existing AI video creation tools and workflows suffer from three significant and interrelated limitations that severely constrain their applicability to complex, narrative-driven production.

The first limitation is functional fragmentation. In current practice, creators must navigate across an array of disconnected models and software tools to complete even a moderately complex video project. A typical workflow might involve employing ChatGPT or Claude for script drafting, Midjourney or Stable Diffusion for character and scene visualization, specialized video generation models for motion synthesis, and conventional non-linear editing software for final assembly and refinement. This fragmented toolchain imposes substantial overhead in terms of context switching, information reformatting, and manual integration, while inevitably introducing information distortion and semantic attenuation at each transfer point between systems. The absence of unified context management means that creative intent is progressively diluted as it passes through the pipeline, with no single component possessing a holistic view of the entire production.

The second limitation is the absence of systematic planning capability. The overwhelming majority of contemporary video generation models are architecturally designed as direct translation engines that map textual prompts to visual outputs through end-to-end differentiable pipelines. While this design enables impressive visual fidelity, it fundamentally lacks any mechanism for narrative reasoning or long-range structural planning. These models cannot comprehend the logical cause-effect relationships that determine why one shot should cinematically follow another, nor can they maintain character consistency across temporal gaps, nor can they deliberately modulate pacing and emotional intensity according to narrative design. Instead, users are forced to rely on increasingly elaborate prompt engineering to exert coarse-grained influence over generation outcomes, an approach that yields results of extreme variability and offers no guarantees of narrative coherence.

The third limitation is prohibitively high correction costs. When a generated output deviates from creative expectations—as frequently occurs given the stochastic nature of the underlying models—the only recourse available to users is to iteratively modify prompts and initiate fresh generation attempts, effectively re-entering the lottery. This process is not only computationally wasteful, consuming substantial GPU resources and

time, but more fundamentally, it offers no mechanism for diagnosing which specific intermediate stage introduced the error. Users cannot interrogate the system to determine whether the problem originated in the script’s structure, the storyboard’s specifications, the visual model’s interpretation, or some combination thereof, and consequently cannot apply targeted corrections. This diagnostic opacity transforms what should be a systematic creative process into an exercise in trial and error.

Taken together, these limitations point to a singular root cause: existing tools and frameworks conceptualize video creation as a single-stage, prompt-to-output generation problem rather than as what it truly is—a planable, executable, and rectifiable long-horizon creative endeavor requiring systematic design, hierarchical decomposition, and iterative refinement. The absence of such a conceptual foundation renders existing systems fundamentally inadequate for the ambitious goal of automated narrative video production.

2.3 Requirements for End-to-End Autonomous Director Agents

Based on the foregoing analysis of task characteristics and existing limitations, this paper proposes that a viable end-to-end autonomous director agent must embody four core capabilities that collectively address the fundamental challenges of complex video creation.

The first capability is task decomposition and strategic planning. The system must possess the ability to receive abstract creative intents expressed in natural language—ranging from a simple story idea to a detailed screenplay—and systematically distill them into executable sub-task sequences organized according to the natural structure of the creative process. This hierarchical decomposition enables the system to manage complexity by addressing manageable subproblems in their appropriate order, with upper-level planning decisions providing constraints rather than specific instructions to lower-level execution modules.

The second capability is specialized multi-agent collaboration. Rather than attempting to imbue a single monolithic model with all the diverse competencies required for video production, the system should employ a division-of-labor architecture in which distinct agents—modeled after the specialized roles of screenwriter, director, cinematographer, and editor in human film production—each assume well-defined functional responsibilities. Through structured inter-agent communication and coordination, the ensemble can address system-level complexity that would overwhelm any individual component, while enabling independent optimization and refinement of each specialized function.

The third capability is comprehensive artifact traceability. The creative process must be externalized through the persistent storage of inspectable intermediate artifacts at every stage—structured scripts, detailed storyboards, material retrieval logs, quality assessment reports, and the dependency relationships linking them. This traceability serves the dual purpose of rendering every creative decision explainable to human users and, equally importantly, enabling precise localization of errors when they occur by tracing backwards through the dependency graph.

The fourth capability is proactive reflection and localized correction. Upon detection of quality issues or deviations from creative intent, the system should not resort to the crude strategy of full regeneration, but rather engage in targeted diagnosis to identify the specific upstream stage responsible for the observed defect, followed by localized repair interventions that address the root cause while preserving the integrity of unaffected components. This capability fundamentally transforms error handling from brute-force retry into precision refinement, with profound implications for both output quality and

computational efficiency.

VibeFilming is systematically designed around these four core requirements, with each architectural decision explicitly motivated by the need to realize one or more of these essential capabilities.

3 VibeFilming Framework Design

3.1 Overall Architecture

VibeFilming adopts a five-stage pipeline architecture organized around a hierarchical collaboration structure. At the apex of this structure sits the “Director Agent,” which functions as the central coordinating hub for the entire system. The Director Agent interfaces with four subordinate specialized agents—the Screenwriter Agent, the Storyboard Agent, the Visual Generation Agent, and the Quality Control Agent—each of which is responsible for a well-defined functional stage of the creative workflow. Upon receiving user input expressed in natural language, which may span the full spectrum from a brief creative premise or a detailed novel synopsis to a fully structured screenplay, the framework automatically initiates the end-to-end production process without requiring further human intervention.

The Director Agent plays a distinctive role within this architecture: it does not directly generate any creative content itself, but instead assumes overarching responsibilities for task allocation across subordinate agents, progress monitoring throughout the pipeline, proactive maintenance of stylistic and thematic consistency across the entire work, and high-stakes exception decision-making when unforeseen issues arise. This role is explicitly modeled on the function of a human director on a professional film set, who coordinates the contributions of specialized department heads without personally executing every technical task.

A critical design principle permeating the entire architecture is that every intermediate artifact produced at each pipeline stage is persistently stored in a structured knowledge base with comprehensive provenance annotations. These annotations capture not only the content of each artifact but also the dependency relationships linking it to preceding and subsequent artifacts, the specific agent that produced it, the timestamp of creation, and any modifications applied through reflection-driven revisions. Together, these stored artifacts constitute a complete creative trajectory from initial concept to final video, ensuring that every decision node is auditable and every error source is traceable.

This design embodies the classical “divide-and-conquer” principle adapted to the specific demands of narrative video production. By decomposing the long-horizon task according to the natural functional divisions observed in human film production, the architecture ensures that each agent confronts only a manageable subset of the overall problem. The substantial reduction in subproblem scale at each stage effectively lowers the cognitive burden and error probability of individual inference steps, while the structured handoffs between stages maintain global coherence.

3.2 Screenwriter Agent: From Creative Concept to Structured Script

The Screenwriter Agent serves as the entry point of the entire creative pipeline, receiving user input and transforming it into a standardized screenplay format. This transformation is of paramount importance, as the quality and structural soundness of the initial script

fundamentally constrain all subsequent stages of the production process—a flawed script cannot be rescued by even the most sophisticated visual generation.

When faced with vague or under-specified creative descriptions, the Screenwriter Agent first invokes a retrieval-augmented generation module that queries both internal knowledge bases and external reference repositories for relevant narrative templates, genre-specific conventions, character archetypes, and scene setting precedents. This retrieval process grounds the generation in established narrative structures, ensuring that the resulting script possesses basic dramatic logic and conforms to audience expectations for the specified genre, whether that be comedy, drama, action, or science fiction.

In cases where the user input consists of a complete novel or extended story, the agent must additionally perform sophisticated narrative compression. This process involves identifying and preserving the essential plot nodes that constitute the narrative backbone, merging or eliminating redundant subplots that would excessively expand the production scope, maintaining key character arcs and developmental trajectories, and ensuring that the compressed narrative retains emotional resonance and dramatic tension despite substantial length reduction. This compression is not a mechanical text summarization but a deliberate distillation guided by narrative importance assessment, with decisions about what to retain and what to prune reflecting judgments about dramatic necessity and production feasibility.

The Screenwriter Agent’s output is a structured script that adheres to professional industry conventions: each scene is numbered sequentially, character dialogues are clearly demarcated from action descriptions, emotional annotations indicate intended tonal qualities, and stage directions provide sufficient detail for subsequent visual interpretation. This structured format ensures that information can be precisely transmitted to downstream agents without ambiguity, while also being machine-parseable for automated processing in the storyboard stage.

3.3 Storyboard Agent: From Script to Visual Blueprint

The Storyboard Agent undertakes the crucial responsibility of translating the textual script into a shootable visual blueprint—a transformation that represents the pivotal transition from abstract narrative to concrete visual design. This stage is arguably the most critical in the entire framework, as it establishes the visual language that will govern all subsequent generation and determines whether the narrative can be effectively rendered as a sequence of compelling images.

The agent employs a recursive narrative decomposition strategy that progressively refines the script across three hierarchical levels. At the highest level, the event level, the agent identifies the core narrative nodes that structure the story—turning points, confrontations, revelations, and resolutions. These events serve as the primary anchors around which the visual plan is organized. At the intermediate level, the scene level, each narrative event is decomposed into executable dramatic units defined by unified spatial and temporal settings, such as a rainy night street encounter or a sunlit interior dialogue. At the finest level, the shot level, the agent generates detailed shooting instructions for each individual camera setup, specifying camera position and angle, shot scale and framing, camera motion trajectories, and compositional requirements including subject placement and depth staging.

This hierarchical planning mechanism is essential for avoiding the combinatorial explosion that would result from attempting to directly generate hundreds of individual

shot specifications from an undifferentiated script. By imposing structure progressively from high-level narrative to low-level visual detail, the system ensures that each stage of planning operates at an appropriate level of abstraction, with upper-level decisions imposing constraints rather than specific implementations on lower levels, enabling top-down refinement while preserving the coherence of the overall narrative vision.

The Storyboard Agent’s output comprehensively documents every scene’s complete shot list with estimated durations, annotated character appearances and positions, key prop requirements, and relevant environmental cues. This document functions as the construction blueprint for the visual generation stage, with its granularity directly determining the executability of subsequent synthesis. Additionally, the agent incorporates a coverage-aware strategy inspired by recent work in AI-assisted video production: throughout the planning process, it actively cross-references available material libraries to identify shots that may lack necessary visual references or asset support, flagging these for proactive resolution rather than allowing them to become blocking issues during generation.

3.4 Visual Generation Agent and Quality Control Agent

The Visual Generation Agent serves as the execution arm of the framework, receiving the storyboard blueprint and invoking underlying video generation models to produce actual visual content. This agent translates the abstract specifications of the plan into concrete visual realizations, and is consequently the stage at which the framework interfaces with the current state of the art in visual synthesis technology.

The agent adopts a two-step “image-first, video-second” strategy that carefully manages the risk of visual inconsistency across shots. In the first step, it generates high-quality keyframes for each shot, including character stills that establish definitive appearance references and scene atmosphere boards that define lighting, color palette, and compositional style. In the second step, these keyframes are used to condition and drive video generation models, enabling the synthesis of dynamic motion while maintaining the established visual baseline. This strategy confers a critical advantage: character appearance consistency, one of the most vexing problems in long-form AI video generation, is locked in at the image stage rather than left to the more volatile video generation process, effectively intercepting the pervasive problem of “character face-switching” before it can emerge.

Throughout the generation process, the Visual Generation Agent maintains an entity anchor library that systematically records the defining visual characteristics of each character, scene, and significant prop that appears in the work. This library provides a shared reference point that ensures visual elements remain stable across shots, even when the same entity appears in widely separated parts of the video with different camera angles and lighting conditions.

The Quality Control Agent operates in parallel with the generation process, performing comprehensive multi-dimensional quality assessment on each generated video clip as it is produced. The agent employs large multi-modal models—architectures capable of joint reasoning about visual content and textual semantics—to assign quantitative scores across dimensions including aesthetic composition, motion coherence, temporal smoothness, character consistency across adjacent shots, and lighting and color matching to the established atmosphere. Each dimension is assessed against both absolute standards of quality and relative consistency with the work’s established visual language.

Crucially, when a clip fails to meet a configured quality threshold, the Quality Control Agent triggers a sophisticated reflection mechanism that fundamentally transforms

the error-handling strategy. Rather than simply instructing for full regeneration—the standard approach in existing systems—the agent first performs diagnostic reasoning to localize the source of the error. The diagnostic process considers multiple hypotheses: were the storyboard instructions insufficiently detailed or ambiguous? Were the required reference materials unavailable in the asset library? Is the error attributable to limitations of the underlying generation model for a particular type of content? Based on this diagnosis, the agent packages feedback that identifies not just the presence of a problem but its likely root cause, then transmits this targeted diagnostic information upstream to the appropriate agent for localized repair. This mechanism upgrades error handling from blind retry to informed revision.

3.5 Director Agent: Coordination, Provenance, and Strategic Decision-Making

The Director Agent occupies the apex of the framework’s organizational hierarchy, functioning as the cognitive and executive center that ensures the entire creative workflow maintains direction and coherence. Its responsibilities span three critical dimensions that collectively determine the system’s ability to deliver a unified, high-quality final product.

At the coordination dimension, the Director Agent actively monitors the progress of each subordinate agent, identifying and resolving contentions and bottlenecks before they can escalate into systemic failures. When multiple shots require the same character assets or scene environments, the agent manages queueing and prioritization to ensure efficient resource utilization. More importantly, the agent maintains vigilance over the work’s aesthetic integrity, promptly intervening when it detects the emergence of stylistic drift—subtle shifts in visual language, color treatment, or cinematic tone that may accumulate across independently generated segments and undermine the work’s unity.

At the provenance dimension, the Director Agent maintains and curates the comprehensive artifact dependency graph that represents the complete creative process. This graph encodes not only the content of each artifact but the relationships connecting them: which upstream decisions generated which downstream specifications, what reasoning justified each major choice, and what alternatives were considered and rejected. This provenance infrastructure serves multiple essential purposes: it enables human users and system designers to understand and audit the system’s creative decisions, it provides the foundation for diagnosing errors by tracing them to their originating stages, and it supports systematic refinement through the identification of recurring failure modes.

At the decision dimension, the Director Agent exercises executive judgment when the Quality Control Agent reports anomalies that require remediation. The agent evaluates repair strategies through a cost-benefit analysis that considers multiple factors: the severity and scope of the detected issue, the availability of alternative pathways, the resource requirements of potential interventions, and the risk that local repairs may introduce secondary problems. Based on this deliberation, the agent may decide to instruct the Storyboard Agent to revise its specifications for a problematic shot series, direct the Visual Generation Agent to retry with different conditioning parameters, or in extreme cases, initiate an alternative narrative pathway that circumvents the problem entirely. This executive capability endows VibeFilming with robust exception-handling abilities that enable it to navigate the inevitable uncertainties and failures of long-horizon creative workflows without catastrophic loss of progress.

4 Experiments and Evaluation

4.1 Experimental Setup

To comprehensively evaluate the VibeFilming framework across varying levels of task complexity, two distinct test scenarios were carefully designed, each targeting different aspects of the system’s capabilities.

Scenario A (short-form narrative) consists of ten creative themes provided to the system as natural language prompts, such as “A cat’s unexpected adventure in outer space” and “The last day of a civilization lost in a forgotten valley.” The system is required to produce narrative shorts of approximately thirty seconds duration comprising between five and eight shots. This scenario primarily assesses the framework’s foundational capabilities in relatively constrained conditions, where the planning horizon is modest and the coordination requirements are manageable.

Scenario B (long-form narrative) employs three short story excerpts, each approximately two thousand Chinese characters in length, with complete narrative arcs including exposition, rising action, climax, and resolution. The system is tasked with generating coherent videos of approximately ninety seconds duration comprising between fifteen and twenty individual shots. This scenario examines the framework’s scalability and robustness in the more demanding context of extended narratives, where planning spans a substantially longer horizon and coordination must maintain coherence across a significantly larger number of segments.

The evaluation employs three baseline methods for comparison. The first is single-model direct generation, using the Seedance 2.5 model driven entirely by carefully engineered prompts without any structural planning or multi-stage refinement. The second and third are two open-source multi-agent frameworks: FilmAgent, a representative system for AI-assisted film production, and a simplified version of ViMax, which omits some of the more sophisticated coordination mechanisms present in the full implementation. These baselines span the spectrum from purely model-based to agent-based approaches, enabling comprehensive and fair comparison across architectural paradigms.

For evaluation, four independent quality dimensions are defined: narrative coherence, measuring the logical consistency and dramatic effectiveness of the story as presented; character consistency, assessing the stability of character appearances, behaviors, and identities across multiple shots; shot smoothness, evaluating the visual and temporal fluidity of transitions between shots; and creative fidelity, gauging the degree to which the final video faithfully realizes the creative intent expressed in the input. Each video is independently scored on a 5-point scale by three evaluators with professional backgrounds in film and television production, with results averaged to mitigate individual subjective bias. Inter-rater reliability was verified through correlation analysis prior to final scoring.

4.2 Main Results

The experimental results, detailed in the accompanying analysis, demonstrate that VibeFilming substantially outperforms all three baseline methods across every evaluation dimension, with the magnitude of improvement varying according to dimension and scenario complexity.

In the short-form Scenario A, VibeFilming achieves a narrative coherence score of 4.2 compared to the single-model baseline’s 2.1, representing an improvement of 2.1 points on the 5-point scale. This substantial margin powerfully validates the central thesis of

this work: the addition of systematic planning and multi-stage refinement yields dramatic improvements in narrative quality that cannot be achieved through even the most sophisticated prompt engineering with a direct generation model. Without a narrative blueprint to guide the process, the single-model approach produces visual content that, while individually plausible, fails to cohere into a meaningful and compelling story.

Against the FilmAgent baseline, VibeFilming shows consistent advantages of approximately 0.6 to 0.8 points across all dimensions. Analysis indicates that these improvements derive primarily from two architectural innovations. First, the Storyboard Agent’s recursive decomposition strategy more effectively maintains narrative continuity across shot transitions, preventing the fragmentation and pacing disruption that frequently afflicts multi-shot generation. Second, the Quality Control Agent’s reflection mechanism substantially reduces the incidence of character inconsistency and visual drift by enabling precise localization and correction of errors as they emerge, rather than allowing them to accumulate unaddressed.

Notably, in Scenario B with its extended ninety-second narratives, the performance gap between VibeFilming and the best baseline widens further, particularly in the dimensions of narrative coherence and character consistency. This scaling behavior suggests that the artifact provenance and dependency management mechanisms at the heart of VibeFilming’s design become increasingly valuable as task complexity increases, delivering architectural leverage that grows with the scope of the production challenge. When errors can be localized precisely and corrected with surgical precision, the system avoids the quality degradation that cumulative errors would otherwise produce over extended sequences.

4.3 Ablation Studies and Discussion

To rigorously validate the independent contribution of each architectural module, systematic ablation experiments were conducted in which individual components were selectively removed or disabled while all other aspects of the system remained unchanged.

Removing the Storyboard Agent entirely—that is, having the Screenwriter Agent directly output a flat, un-hierarchical list of shot specifications without the intervening recursive decomposition stage—caused narrative coherence scores to decline substantially from 4.2 to 3.5. This result indicates that the hierarchical planning provided by the Storyboard Agent is not merely a convenience but an essential requirement for maintaining logical self-consistency in any narrative of meaningful length. Direct translation from script to shot list systematically weakens the causal and dramatic relationships between shots, producing a sequence that feels episodic and arbitrary rather than organically flowing.

Disabling the Quality Control Agent’s reflection mechanism and reverting to the standard practice of full regeneration upon error detection dramatically increased the average generation time for Scenario B by a factor of 3.7, while yielding no detectable improvement in final output quality. This result provides definitive empirical evidence for the efficiency advantage of localized repair over brute-force global retry, with the difference becoming more pronounced as task length increases. Each full regeneration discards all progress made to date, including correctly functioning components, while the reflection mechanism preserves what works and intervenes only where needed.

Removing the Director Agent’s coordination function—allowing each subordinate agent to operate independently without centralized oversight—produced a marked decline in

cross-shot visual consistency from 4.3 to 3.6. Investigation of the generated outputs revealed that in the absence of a shared aesthetic anchor and unified scheduling constraints, the independent decisions of the various agents progressively diverged, accumulating into noticeable style drift across the video. This finding underscores the essential role of centralized coordination in multi-agent creative systems, where the whole must actively manage the parts to prevent the emergent fragmentation that independent optimization naturally produces.

In the discussion dimension, several important qualifications merit attention. First, VibeFilming’s output quality remains fundamentally constrained by the representational capacity of the underlying video generation models. For shots requiring complex physical interactions—such as fluid dynamics, object collisions, or articulated character movements—or for scenes demanding subtle facial expressions that convey emotional nuance, even perfect planning and quality control cannot fully compensate for generative model limitations. This confirms the prevailing industry consensus that the quality ceiling for automated video production is ultimately determined by generative model capabilities, with the agent framework’s value lying in efficiently approaching that ceiling while providing controllability and interpretability.

Second, the current framework entails substantial inference costs. A complete generation workflow for a ninety-second video requires dozens of calls to various large language and multi-modal models, consuming significant computational resources and API tokens per production attempt. While the cost is acceptable for professional production scenarios, it remains a substantial barrier to interactive or iterative creative workflows where users may wish to experiment with multiple variations. Optimizing the efficiency of inter-agent communication and reducing the number of expensive model calls represents a key direction for future work.

5 Conclusion and Future Work

This paper has presented VibeFilming, an end-to-end autonomous director agent framework designed to address the fundamental challenges of complex, narrative-driven video creation through systematic planning, specialized multi-agent collaboration, and adaptive error recovery. The contributions of this work are threefold and mutually reinforcing.

First, the paper reconceptualizes video creation as fundamentally a planable, executable, and rectifiable long-horizon task rather than the single-shot, prompt-to-output generation problem that has dominated the field. This perspective shift reframes the research agenda from improving isolated generative models to designing comprehensive systems capable of managing the full creative lifecycle.

Second, the paper introduces a five-stage multi-agent architecture in which each specialized agent—Screenwriter, Storyboard, Visual Generation, Quality Control, and Director—assumes clearly defined professional functions that closely correspond to the division of labor observed in human film production. This structural decomposition enables systematic complexity management at the architectural level, with each agent optimized for its specific domain while the Director Agent maintains overall coherence.

Third, the paper contributes artifact provenance tracking and reflection mechanisms that transform error handling from blind regeneration into informed, localized repair. By externalizing the creative process as inspectable intermediate artifacts and maintaining the dependency relationships among them, the framework enables precise diagnosis of error sources and targeted interventions that preserve unaffected work product while

addressing identified deficiencies.

Experimental validation across both short-form and long-form narrative scenarios demonstrates that VibeFilming significantly outperforms existing single-model and multi-agent baselines in narrative coherence, character consistency, shot smoothness, and creative fidelity. Ablation studies confirm the essential contributions of each architectural component and provide empirical evidence for the efficiency advantages of localized error repair over full regeneration.

Looking toward the future, three directions hold particular promise for extending and enhancing the VibeFilming framework. First, the development of more efficient inter-agent communication protocols and collaboration mechanisms could substantially reduce inference costs, making the framework accessible for interactive creative workflows and resource-constrained deployment scenarios. Second, deep integration with next-generation long-video generation models—such as systems offering native generation of extended sequences with robust temporal consistency—could leverage improvements in generative capacity to raise the quality ceiling of the entire system. Third, large-scale user studies and field validation conducted in collaboration with professional filmmakers and production studios will provide essential feedback about the framework’s utility, usability, and adaptability in authentic creative workflows, informing iterative refinement of both the architecture and its interaction design.

References

- [1] Xu Z, Wang J, Wang L, et al. FilmAgent: Automating Virtual Film Production Through a Multi-Agent Collaborative Framework. In: Proceedings of SIGGRAPH Asia 2024 Technical Communications. ACM, 2024, pp. 1–4.
- [2] Kling Team. Kling-Omni: Omni Video Generation Model from Multi-modal Visual Language. arXiv preprint, 2025.
- [3] Multi-University Joint Team. UniVA: A General AI Video Agent Framework. AI Community, 2025.
- [4] Huang C Team, University of Hong Kong. ViMax: Agentic Video Generation Framework. GitHub Repository, 2025.
- [5] Crayotter Team. Crayotter: Traceable Multi-Agent Workflows for Long-Form Video Editing. arXiv preprint, 2026.
- [6] Volcano Engine. Seedance 2.5 Video Generation Model Technical Report. 2026.
- [7] iQIYI. Nadou Pro Professional Film Production Platform Technical White Paper. 2026.